

final project

formulate a novel RL problem and solve it

tl;dr

20 points

group of size 1 to 6

submit the following

- 1 submission per group online
- project **report** (ppt / pdf), no more than 20 pages
- **videos** of the agent “playing” the game
- **source code** (zip)
 - but not the model checkpoints as they could be too large

grading criteria

RL formulation (10 pts)

- **novelty** the RL formulation should be your own and novel (5 pts)
- **formalism** clearly state the MDP (or POMDP) formulation (2 pts)
- **env** the MDP is implemented in a gymnasium style environment code (3 pts)

RL solution (10 pts)

- **showcase** videos of agents “playing” the game (5 pts)
 - **document** the training process of the agent (5 pts)
- report
 - videos
 - source code

report slide template

- title page
- **group members** ← important
- RL game description and formulation
- RL game solution

make the slides clean and to the point



Language & program abstractions

Programs are compositional. This compositional structure is reflected in language!

*pick up a knife, find a tomato, then
go to the counter and slice it*

```
find_knife
look(down), forward(6), rotate(90), forward(17),
    find_tomato
forward(3), rotate(90), look(down), pick(obj1)
goto_counter
rotate(90), forward(2), rotate(270), forward(1),
    slice
forward(1), look(down), cut(obj2, obj1), look(up),
```

*a pinwheel made
of 8 heptagons*

```
pinwheel(8,
gon(7, 1cm))
```



*a pinwheel made
of 5 pentagons*

```
pinwheel(5,
gon(5, 2cm))
```



good

Creating a good presentation involves several key steps and considerations, from planning and organizing your content to delivering the presentation effectively. Here's a step-by-step guide on how to make a good presentation:

❖ Define Your Purpose and Audience

- Start by clearly defining the purpose of your presentation. What do you want to achieve with it? Who is your target audience? Understanding your objectives and audience will help shape your content and approach.

❖ Research and Gather Information

- Conduct thorough research on your topic to ensure you have accurate and up-to-date information.
- Organize your content logically, breaking it down into key points, & supporting details.

❖ Create an Outline

- Develop an outline that outlines the structure of your presentation. Start with an introduction, followed by the main content, and conclude with a summary or call to action.

❖ Design Engaging Slides

- Use a clean and consistent design for your slides. Choose a readable font and maintain a uniform color scheme.
- Use visuals such as images, charts, and graphs to enhance understanding and engagement.
- Limit the amount of text on each slide. Use bullet points or short rather than lengthy paragraphs.

❖ Tell a Story

- Craft your presentation as a narrative that flows logically from one point to the next. This helps your audience follow along and stay engaged.
- Use anecdotes, examples, and real-life stories to make your content relatable.

❖ Practice and Rehearse:

- Practice your presentation multiple times to become familiar with the content and the flow.
- Pay attention to your timing to ensure you stay within your allotted presentation time.

❖ Engage Your Audience:

- Start with a compelling hook.
- Encourage interaction through polls and surveys if applicable.
- Use eye contact and body language.

❖ Use Visual Aids Effectively

- Use slides as visual aids.
- Use visuals to reinforce key points.

❖ Practice Good Delivery

- Speak clearly and at a moderate pace.
- Vary your tone and pace.
- Minimize the use of filler words.

❖ Handle Questions and Comments

- Prepare for questions and comments.
- Be open to feedback and constructive criticism.

❖ Summarize and Conclude

- End your presentation with a strong conclusion.
- Conclude with a strong closing statement.

❖ Gather Feedback

- After your presentation, seek feedback from your audience.
- Review your presentation and your own assessment.

bad

novelty

the RL “game” should be something you came up with, rather than using existing RL environments such as those from <https://gymnasium.farama.org/>

it should also not be derived from asking claude “give me a novel RL task”

- something like the elevator formulation from week 1 is a good candidate
- don't make it too hard (as we know RL is difficult)

formalism

describe the MDP's state/action/transition/reward

estimate of the size of all states, actions

env

you should have [env.py](#) in your submitted source code folder

video

- not all RL tasks are “game-like”, but you need to demonstrate the agent taking a sequence of decisions
- the following videos are required (but not limited to):
 - the random agent playing the game (it should be clear that it is bad at the game)
 - the final trained agent playing the game (it should appear better at the game)

training process

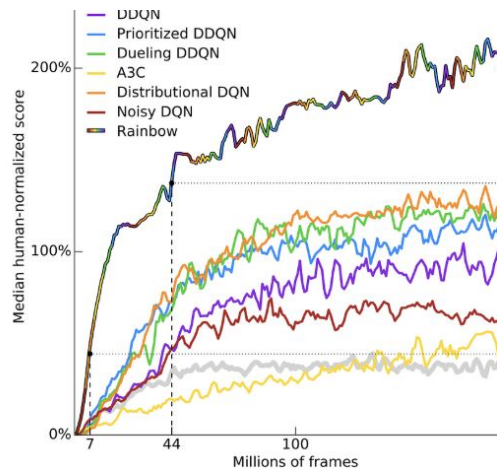
explain the RL algorithms used to play the game, along with additional tricks like reward shaping etc . . .

a plot of the training curve of a few training runs:

- x axis: number of environment steps taken
- y axis: raw reward obtained from the game

don't stress too hard if RL fails to train (it happens)

- but in the report explain why it failed to train



something like this